

Fourier Towers for Higher Nilpotent Symmetry: Nested Cocycle Layers in $UT_4(\mathbb{Z}_5)$

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Abstract

We develop a Fourier Tower interpretation of the finite nilpotent group

$$UT_4(\mathbb{Z}_5),$$

the group of upper unitriangular 4×4 matrices over $\mathbb{Z}_5$.

The motivation is that $UT_3(\mathbb{Z}_5)$ is exactly the finite Heisenberg group already studied in the Fourier Tower framework. In that case, nonabelianity is encoded by a single central cocycle, which becomes a controlled phase interaction after Fourier transformation.

The group $UT_4(\mathbb{Z}_5)$ presents a richer structure: it has several nilpotent layers and multiple cocycle interactions. It therefore provides a natural higher-order example of the Fourier Tower philosophy.

The central idea is that nonabelian structure may be decomposed into:

local Fourier layers + cocycle phase interactions + higher nilpotent propagation.

We compare $UT_4(\mathbb{Z}_5)$ with the finite Heisenberg group and with affine groups. The comparison shows that Heisenberg groups model central phase geometry, affine groups model semidirect permutation geometry, while $UT_4(\mathbb{Z}_5)$ combines several nested cocycle layers.

This work is exploratory and aims to provide a structured language connecting finite-group Fourier analysis, nilpotent groups, cocycle geometry, and quantum circuit interpretations.

1 Introduction

Fourier analysis on finite groups plays an important role in harmonic analysis, representation theory, and quantum computation.

For finite abelian groups, Fourier analysis decomposes functions into independent characters. For nonabelian groups, the structure becomes more complicated because irreducible representations are no longer one-dimensional.

The Fourier Tower viewpoint proposes an intermediate approach for solvable and nilpotent groups.

Instead of diagonalizing the full nonabelian group algebra at once, one starts from a subgroup tower

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$$

with abelian quotient layers

$$Q_i = G_i/G_{i-1}.$$

Each quotient layer is Fourier transformed locally, and the nonabelian structure appears through interactions between these layers.

The guiding principle is:

$$\boxed{\text{nonabelianity} = \text{interaction between local Fourier layers.}}$$

In the finite Heisenberg case

$$UT_3(\mathbb{Z}_5),$$

there is one main cocycle interaction. This case has already been developed as the basic model of cocycle-induced Fourier phase geometry.

The purpose of the present article is to go one step higher and study

$$UT_4(\mathbb{Z}_5).$$

This group is more interesting for the Fourier Tower framework because it contains:

- first-order abelian coordinates,
- second-order cocycle coordinates,
- a third-order central coordinate,
- and nested phase interactions.

Thus $UT_4(\mathbb{Z}_5)$ provides a natural example of a higher nilpotent Fourier Tower.

2 General Fourier Tower Philosophy

Let G be a finite solvable group with a tower

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G.$$

Assume the quotients

$$Q_i = G_i/G_{i-1}$$

are abelian.

To each quotient layer one associates the Hilbert space

$$\mathcal{H}_i = \mathbb{C}[Q_i].$$

The local Fourier layer is

$$F_{\text{loc}} = \bigotimes_i F_{Q_i}.$$

The nonabelian information is not contained in the local Fourier transforms alone. It appears through the way the layers interact.

These interactions may be:

- controlled permutations,
- semidirect-product actions,
- cocycle-induced phases,
- central extension phases,
- or higher nilpotent corrections.

Thus a Fourier Tower transform has the schematic form

$$F_{\mathcal{T}} = C_{\text{tower}} F_{\text{loc}},$$

where C_{tower} contains the coupling data between quotient Fourier layers.

3 Reminder: $UT_3(\mathbb{Z}_5)$ is the Finite Heisenberg Group

Before studying $UT_4(\mathbb{Z}_5)$, we recall that

$$UT_3(\mathbb{Z}_5)$$

is exactly the finite Heisenberg group over $\mathbb{Z}_5$.

Its elements are

$$\begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \quad a, b, c \in \mathbb{Z}_5.$$

We identify such a matrix with a triple

$$(a, b, c).$$

The multiplication law is

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab').$$

The term

$$ab'$$

is the cocycle responsible for noncommutativity.

The center is

$$Z(UT_3(\mathbb{Z}_5)) = \{(0, 0, c) : c \in \mathbb{Z}_5\} \cong \mathbb{Z}_5.$$

The quotient is

$$UT_3(\mathbb{Z}_5)/Z \cong \mathbb{Z}_5^2.$$

Thus

$$UT_3(\mathbb{Z}_5) \cong (\mathbb{Z}_5^2) \times_{\omega} \mathbb{Z}_5,$$

with cocycle

$$\omega((a, b), (a', b')) = ab'.$$

After Fourier transformation of the central coordinate, translation in c becomes a phase.

Thus the cocycle becomes a controlled phase interaction:

$$\omega_5^{ab'}, \quad \omega_5 = e^{2\pi i/5}.$$

This is the basic Heisenberg Fourier Tower already studied.

In this sense:

$$\boxed{UT_3(\mathbb{Z}_5) = \text{one cocycle layer.}}$$

4 The Group $UT_4(\mathbb{Z}_5)$

The group

$$UT_4(\mathbb{Z}_5)$$

consists of matrices

$$g(a, b, c, d, e, f) = \begin{pmatrix} 1 & a & b & c \\ 0 & 1 & d & e \\ 0 & 0 & 1 & f \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

where

$$a, b, c, d, e, f \in \mathbb{Z}_5.$$

Hence

$$|UT_4(\mathbb{Z}_5)| = 5^6.$$

We use the coordinates

$$(a, d, f)$$

for the first superdiagonal,

$$(b, e)$$

for the second superdiagonal, and

$$c$$

for the top central coordinate.

Thus the matrix is organized as:

$$\begin{array}{cccc} 1 & a & b & c \\ 0 & 1 & d & e \\ 0 & 0 & 1 & f \\ 0 & 0 & 0 & 1. \end{array}$$

4.1 Multiplication Law

Let

$$g = (a, b, c, d, e, f)$$

and

$$g' = (a', b', c', d', e', f').$$

Then

$$gg' = (A, B, C, D, E, F),$$

where

$$A = a + a',$$

$$D = d + d',$$

$$F = f + f',$$

$$B = b + b' + ad',$$

$$E = e + e' + df',$$

and

$$C = c + c' + ae' + bf'.$$

Thus:

$$\begin{aligned} & (a, b, c, d, e, f)(a', b', c', d', e', f') \\ &= (a + a', b + b' + ad', c + c' + ae' + bf', d + d', e + e' + df', f + f'). \end{aligned}$$

This multiplication law displays several layers of interaction.

The first-order coordinates are:

$$a, d, f.$$

The second-order cocycle coordinates are:

$$b, e.$$

The third-order central coordinate is:

$$c.$$

The cocycle terms are:

$$ad', \quad df', \quad ae', \quad bf'.$$

Thus $UT_4(\mathbb{Z}_5)$ has multiple nested cocycle interactions.

5 Nilpotent Tower Structure

The group $UT_4(\mathbb{Z}_5)$ is nilpotent.

Its lower central series has the form

$$UT_4(\mathbb{Z}_5) = \Gamma_1 \triangleright \Gamma_2 \triangleright \Gamma_3 \triangleright \Gamma_4 = \{e\}.$$

Here:

$$\Gamma_2 = [\Gamma_1, \Gamma_1]$$

consists of matrices with zero first superdiagonal:

$$\Gamma_2 = \left\{ \begin{pmatrix} 1 & 0 & b & c \\ 0 & 1 & 0 & e \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} : b, c, e \in \mathbb{Z}_5 \right\}.$$

The next subgroup is

$$\Gamma_3 = [\Gamma_1, \Gamma_2],$$

which is the center:

$$\Gamma_3 = Z(UT_4(\mathbb{Z}_5)) = \left\{ \begin{pmatrix} 1 & 0 & 0 & c \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} : c \in \mathbb{Z}_5 \right\}.$$

Therefore the quotient layers are:

$$\Gamma_1/\Gamma_2 \cong \mathbb{Z}_5^3,$$

$$\Gamma_2/\Gamma_3 \cong \mathbb{Z}_5^2,$$

and

$$\Gamma_3 \cong \mathbb{Z}_5.$$

Thus the Fourier Tower has three main layers:

$$\mathbb{Z}_5^3, \quad \mathbb{Z}_5^2, \quad \mathbb{Z}_5.$$

This is richer than the Heisenberg case, which only has:

$$\mathbb{Z}_5^2 \quad \text{and} \quad \mathbb{Z}_5.$$

6 Fourier Layers for $UT_4(\mathbb{Z}_5)$

The natural Hilbert space is

$$\mathcal{H} = \mathbb{C}[UT_4(\mathbb{Z}_5)].$$

Using the tower coordinates, we may identify it as

$$\mathcal{H} \cong \mathbb{C}[\mathbb{Z}_5^3] \otimes \mathbb{C}[\mathbb{Z}_5^2] \otimes \mathbb{C}[\mathbb{Z}_5].$$

Equivalently,

$$\mathcal{H} \cong \mathbb{C}[\mathbb{Z}_5]^{\otimes 6}.$$

The first layer corresponds to:

$$(a, d, f) \in \mathbb{Z}_5^3.$$

The second layer corresponds to:

$$(b, e) \in \mathbb{Z}_5^2.$$

The central layer corresponds to:

$$c \in \mathbb{Z}_5.$$

Let

$$\omega_5 = e^{2\pi i/5}.$$

The Fourier transform on $\mathbb{Z}_5$ is

$$F_5|x\rangle = \frac{1}{\sqrt{5}} \sum_{k=0}^4 \omega_5^{kx} |k\rangle.$$

Therefore the local Fourier transform is

$$F_{\text{loc}} = F_5^{\otimes 6}.$$

More structurally,

$$F_{\text{loc}} = F_{\mathbb{Z}_5^3} \otimes F_{\mathbb{Z}_5^2} \otimes F_{\mathbb{Z}_5}.$$

7 Cocycle Interactions

The multiplication law contains the nested cocycle corrections.

Although the quotient layers

$$\Gamma_i/\Gamma_{i+1}$$

do not themselves form a single exact sequence, each step of the lower central tower defines an exact extension sequence

$$1 \rightarrow \Gamma_{i+1} \rightarrow \Gamma_i \rightarrow \Gamma_i/\Gamma_{i+1} \rightarrow 1.$$

The nontriviality of these extensions is encoded by cocycle corrections appearing in the multiplication law.

From the Fourier Tower viewpoint, these cocycles generate the phase interactions coupling the local Fourier layers.

$$b \mapsto b + ad',$$

$$e \mapsto e + df',$$

and

$$c \mapsto c + ae' + bf'.$$

After Fourier transformation of the (b, e, c) -layers, these translations become phase factors. Let the Fourier dual variables of b, e, c be denoted by:

$$\beta, \epsilon, \gamma \in \mathbb{Z}_5.$$

Then the translations become phases:

$$b \mapsto b + ad' \rightsquigarrow \omega_5^{\beta ad'},$$

$$e \mapsto e + df' \rightsquigarrow \omega_5^{\epsilon df'},$$

and

$$c \mapsto c + ae' + bf' \rightsquigarrow \omega_5^{\gamma(ae' + bf')}.$$

Thus the full cocycle phase has the schematic form:

$$\omega_5^{\beta ad' + \epsilon df' + \gamma(ae' + bf')}.$$

This is the key Fourier Tower feature of $UT_4(\mathbb{Z}_5)$.

It has several interacting phase layers instead of a single Heisenberg phase.

Theorem 1 (Schematic Fourier Tower Decomposition for $UT_4(\mathbb{Z}_5)$). *The group $UT_4(\mathbb{Z}_5)$ admits a schematic Fourier Tower decomposition with quotient layers*

$$\mathbb{Z}_5^3, \quad \mathbb{Z}_5^2, \quad \mathbb{Z}_5.$$

After local Fourier transformation of these abelian layers, the nilpotent multiplication law is represented by nested cocycle-controlled phase interactions.

Schematically,

$$F_{\mathcal{T}}^{UT_4} = C_{\omega}^{(3)} C_{\omega}^{(2)} F_{\text{loc}},$$

where

$$F_{\text{loc}} = F_{\mathbb{Z}_5^3} \otimes F_{\mathbb{Z}_5^2} \otimes F_{\mathbb{Z}_5},$$

and the operators

$$C_{\omega}^{(2)} \quad \text{and} \quad C_{\omega}^{(3)}$$

encode respectively the second-order and third-order cocycle interactions.

Thus the nonabelian structure of $UT_4(\mathbb{Z}_5)$ appears in Fourier space as a hierarchy of phase couplings.

Proof. The lower central series gives the abelian quotient layers

$$\Gamma_1/\Gamma_2 \cong \mathbb{Z}_5^3,$$

$$\Gamma_2/\Gamma_3 \cong \mathbb{Z}_5^2,$$

and

$$\Gamma_3 \cong \mathbb{Z}_5.$$

Each layer therefore admits an ordinary abelian Fourier transform.

The multiplication law of $UT_4(\mathbb{Z}_5)$ contains the correction terms

$$ad', \quad df', \quad ae' + bf'.$$

These terms translate the higher nilpotent coordinates b, e, c .

Fourier transformation converts translations in these coordinates into phase multiplications.

Therefore the nilpotent correction terms become diagonal cocycle-controlled phase gates.

Hence the Fourier Tower transform may be represented schematically as a local abelian Fourier layer followed by nested cocycle phase interactions. □

The primed variables should be understood as external multiplication variables or as control data coming from a second group-register in a full multiplication circuit. The diagram below is therefore schematic: it displays the cocycle phase structure rather than a complete two-register group multiplication circuit.

8 Circuit Interpretation

The Fourier Tower circuit for $UT_4(\mathbb{Z}_5)$ contains six qudit registers:

$$|a\rangle, |d\rangle, |f\rangle, |b\rangle, |e\rangle, |c\rangle.$$

The first three represent the first superdiagonal layer:

$$(a, d, f).$$

The next two represent the second superdiagonal layer:

$$(b, e).$$

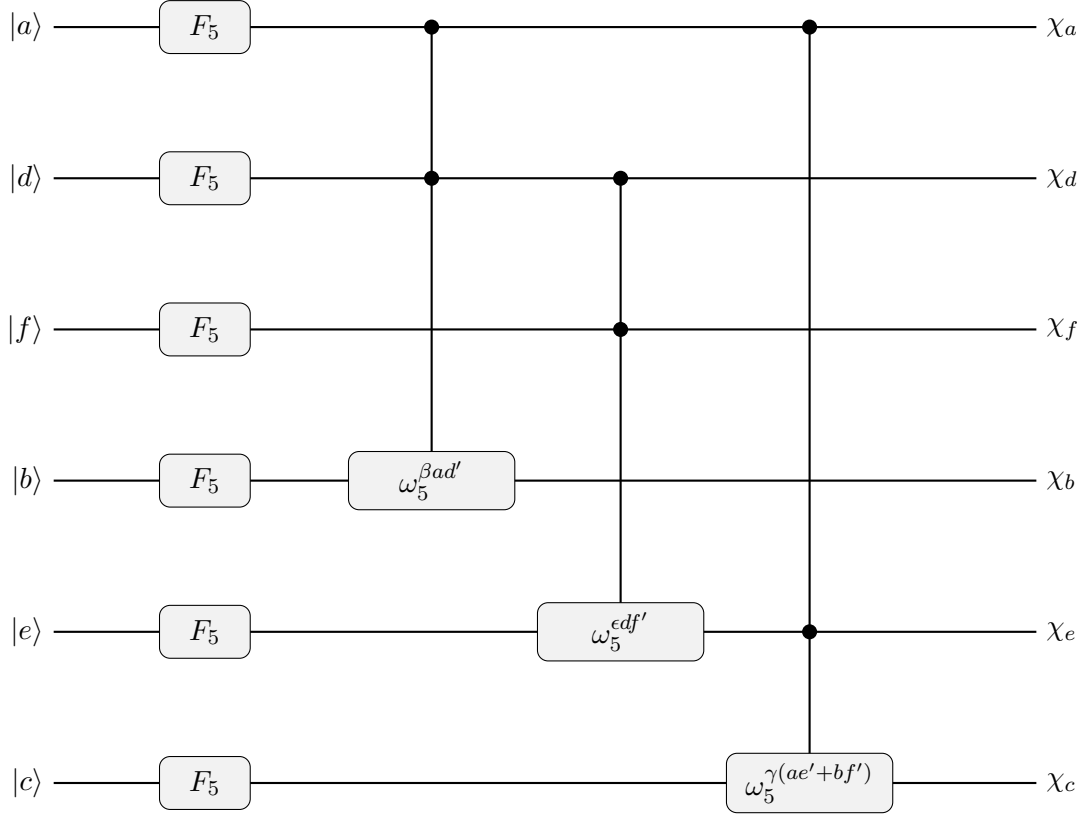
The last one represents the central layer:

$$c.$$

The circuit has the following stages:

local Fourier transforms $\longrightarrow$ second-order cocycle phases $\longrightarrow$ third-order central phases.

A schematic circuit is:



This diagram is schematic and represents the cocycle phase geometry associated with group multiplication.

The primed variables appearing in the phase factors arise from a second group element in a full multiplication representation

$$|g\rangle|g'\rangle.$$

Accordingly, the diagram should be interpreted as displaying the hierarchy of cocycle-controlled phase interactions rather than a complete gate-level multiplication circuit. Its purpose is to show the hierarchy of Fourier layers and cocycle-controlled phase interactions.

9 Comparison with the Heisenberg Group

The finite Heisenberg group

$$UT_3(\mathbb{Z}_5)$$

has the structure:

$$\mathbb{Z}_5^2 \longrightarrow \mathbb{Z}_5.$$

It has one central cocycle:

$$ab'.$$

Its Fourier Tower structure is:

two abelian coordinates + one central phase.

By contrast,

$$UT_4(\mathbb{Z}_5)$$

has the structure:

$$\mathbb{Z}_5^3 \longrightarrow \mathbb{Z}_5^2 \longrightarrow \mathbb{Z}_5.$$

It has several nested cocycle terms:

$$ad', \quad df', \quad ae' + bf'.$$

Thus:

$$UT_3(\mathbb{Z}_5) = \text{single cocycle Fourier Tower},$$

while

$$UT_4(\mathbb{Z}_5) = \text{multi-layer cocycle Fourier Tower}.$$

This makes $UT_4(\mathbb{Z}_5)$ a more powerful example for demonstrating the hierarchical nature of Fourier Towers.

10 Comparison with Affine Groups

Affine groups provide a different type of nonabelian structure.

For example, the affine group over $\mathbb{Z}_5$ is

$$\text{Aff}(\mathbb{Z}_5) = \mathbb{Z}_5 \rtimes \mathbb{Z}_5^\times.$$

Its elements act by

$$x \mapsto ax + b,$$

where

$$a \in \mathbb{Z}_5^\times, \quad b \in \mathbb{Z}_5.$$

Here the nonabelian structure comes from a semidirect product:

$$(a, b)(a', b') = (aa', b + ab').$$

This is different from the nilpotent cocycle structure of $UT_4(\mathbb{Z}_5)$.

In the affine case:

$$\text{nonabelianity} = \text{action of one layer on another.}$$

In the Heisenberg case:

$$\text{nonabelianity} = \text{central cocycle phase.}$$

In the $UT_4(\mathbb{Z}_5)$ case:

$$\text{nonabelianity} = \text{nested cocycle phase geometry.}$$

Thus the Fourier Tower framework separates three important mechanisms:

Affine groups	: semidirect permutation/action layers,
Heisenberg groups	: single central cocycle phase,
$UT_4(\mathbb{Z}_5)$	: multiple nested cocycle phases.

This comparison shows why $UT_4(\mathbb{Z}_5)$ is a natural next example after the Heisenberg group.

11 Interpretation

The group $UT_4(\mathbb{Z}_5)$ suggests the following hierarchy:

local Fourier layers $\longrightarrow$ secondary cocycle phases $\longrightarrow$ central higher-order phases.

This is precisely the type of structure the Fourier Tower framework is designed to describe.

The Heisenberg group shows how one cocycle becomes one phase interaction.

The group $UT_4(\mathbb{Z}_5)$ shows how several cocycles can propagate through a tower and produce a hierarchy of phase interactions.

This suggests a broader principle:

nilpotent groups = iterated cocycle Fourier geometries.

11.1 Intertwining Interpretation

The Fourier Tower transform may also be viewed as a type of intertwining operator between local abelian Fourier sectors and the global nilpotent group action.

Indeed, the local Fourier transform

$$F_{\text{loc}}$$

decomposes the quotient layers into independent harmonic coordinates, while the cocycle phase operators reconnect these layers through the nonabelian multiplication law.

From this viewpoint, the Fourier Tower transform may be represented schematically as

$$F_{\mathcal{T}} = C_{\omega} F_{\text{loc}},$$

acting as a twisted intertwiner between:

- local Fourier representations of the quotient layers,
- and the global representation structure of $UT_4(\mathbb{Z}_5)$.

This interpretation suggests possible connections with induced representations, Mackey-type constructions, and harmonic analysis on nilpotent groups.

12 Conclusion

The group $UT_4(\mathbb{Z}_5)$ provides a richer Fourier Tower example than the finite Heisenberg group.

While

$$UT_3(\mathbb{Z}_5)$$

is already the finite Heisenberg group and has one central cocycle, the group

$$UT_4(\mathbb{Z}_5)$$

contains several nilpotent layers and multiple nested cocycle interactions.

This makes it a natural example for developing higher Fourier Towers.

The main conceptual picture is:

nonabelianity = nested interaction between local Fourier layers.

Future work may include:

- explicit matrix computations of the Fourier Tower transform for $UT_4(\mathbb{Z}_5)$,
- extensions to $UT_n(\mathbb{Z}_p)$,
- connections with nilpotent harmonic analysis,
- and possible quantum circuit interpretations of higher cocycle phase gates.

The constructions presented here are primarily schematic and conceptual; the goal is to propose an organizational viewpoint relating nilpotent group structure, cocycle extensions, and layered Fourier interactions rather than a complete representation-theoretic classification.

References

- [1] P. Diaconis, *Group Representations in Probability and Statistics*, Institute of Mathematical Statistics, 1988.
- [2] A. Terras, *Fourier Analysis on Finite Groups and Applications*, Cambridge University Press, 1999.
- [3] J.-P. Serre, *Linear Representations of Finite Groups*, Springer, 1977.
- [4] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.
- [5] R. Howe, “On the Role of the Heisenberg Group in Harmonic Analysis,” *Bulletin of the American Mathematical Society*, Vol. 3, No. 2, 1980.
- [6] M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2000.
- [7] D. Gottesman, *Stabilizer Codes and Quantum Error Correction*, PhD thesis, Caltech, 1997.
- [8] S. Aaronson and D. Gottesman, “Improved Simulation of Stabilizer Circuits,” *Physical Review A*, 70, 052328 (2004).